

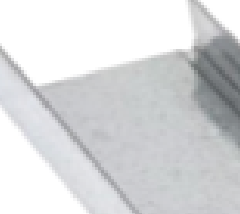


PRODUCT

GRADE

GAUGE

COATING



PUBLISHED PERFORMANCE DATA

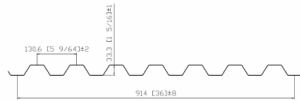
1316FD-FY50-20-G90 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

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GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (lb/ft)	E, ksi	S _x (in ³ /ft)	S _y (in ³ /ft)	ASD (0 + 16%) M _u / 0 (in ³ -lbs/ft)	ASD (0 + 16%) M _u / 0 (in ³ -lbs/ft)	I _x (in ⁴ /ft)	I _y (in ⁴ /ft)
22	0.0096	16	50	0.179	0.170	5308	508	0.107	0.144
20	0.0098	20	50	0.222	0.216	6681	6407	0.210	0.182
18	0.0099	24	50	0.300	0.294	9081	8802	0.290	0.257
16	0.0098	30	50	0.390	0.378	11667	11327	0.363	0.341

Note: All section properties and ASD flexural strengths are calculated in accordance with AISI/S430-2017, AISI S430-2017 and AISI S430-2018.

Shear and Web Crippling

Gage	V _u (lb/ft)	Web Crippling (R _u) One Flange Loading End Bearing	Web Crippling (R _u) One Flange Loading Interior Bearing
22	2424	1-1/2" 801 2" 880 3" 1015	1-1/2" 1096 2" 1189 3" 1345
20	3053	1-1/2" 1053 2" 1155 3" 1306	1-1/2" 1395 2" 1500 3" 1653
18	5032	1-1/2" 2087 2" 2377 3" 2773	1-1/2" 2983 2" 3354 3" 3854
16	6279	1-1/2" 2591 2" 2983 3" 3406	1-1/2" 4240 2" 4694 3" 5272

Note: All section properties and ASD flexural strengths are calculated in accordance with AISI/S430-2017, AISI S430-2017 and AISI S430-2018.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
Single	22	143	138	133	128	123	118	113	108	103
	20	178	173	168	163	158	153	148	143	138
	18	213	208	203	198	193	188	183	178	173
	16	248	243	238	233	228	223	218	213	208
Double	22	138	133	128	123	118	113	108	103	98
	20	173	168	163	158	153	148	143	138	133
	18	208	203	198	193	188	183	178	173	168
	16	243	238	233	228	223	218	213	208	203
Triple	22	133	128	123	118	113	108	103	98	93
	20	168	163	158	153	148	143	138	133	128
	18	203	198	193	188	183	178	173	168	163
	16	238	233	228	223	218	213	208	203	198

Note: 1. All section properties and ASD (0 + 16%) uniform loads are calculated in accordance with AISI/S430-2017, AISI S430-2017 and AISI S430-2018.
2. Loads shown in tables are uniformly distributed superimposed loads in psf. Span length assumes center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
3. Bending Moment (inches used for design) shall be Single and Two Span M_u = 8' and 10' respectively.
4. Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span condition.

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Span	Gage	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
Single	22	68	63	58	53	48	43	38	33	28
	20	86	81	76	71	66	61	56	51	46
	18	104	99	94	89	84	79	74	69	64
	16	122	117	112	107	102	97	92	87	82
Double	22	54	49	44	39	34	29	24	19	14
	20	68	63	58	53	48	43	38	33	28
	18	82	77	72	67	62	57	52	47	42
	16	96	91	86	81	76	71	66	61	56
Triple	22	49	44	39	34	29	24	19	14	9
	20	63	58	53	48	43	38	33	28	23
	18	77	72	67	62	57	52	47	42	37
	16	91	86	81	76	71	66	61	56	51

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.67.

Construction Span Table – 20 psf Construction Load

Normal Weight Concrete (150 pcf)					Lightweight Concrete (100 pcf)				
Total Slab Depth	Deck Type	Maximum Unshored Clear Span			Total Slab Depth	Deck Type	Maximum Unshored Clear Span		
		1 span	2 span	3 span			1 span	2 span	3 span
4'-0" (R200) 40 pcf	15x6x20 ga	7'-0"	6'-0"	5'-0"	4'-0" (R200) 40 pcf	15x6x20 ga	8'-0"	6'-0"	5'-0"
	15x6x18 ga	7'-0"	6'-0"	5'-0"		15x6x18 ga	8'-0"	7'-0"	6'-0"
	15x6x16 ga	6'-0"	5'-0"	4'-0"		15x6x16 ga	7'-0"	6'-0"	5'-0"
	15x6x14 ga	6'-0"	5'-0"	4'-0"		15x6x14 ga	6'-0"	5'-0"	4'-0"
	15x6x12 ga	5'-0"	4'-0"	3'-0"		15x6x12 ga	5'-0"	4'-0"	3'-0"
4'-0" (R200) 42 pcf	15x6x20 ga	7'-0"	6'-0"	5'-0"	4'-0" (R200) 42 pcf	15x6x20 ga	7'-0"	6'-0"	5'-0"
	15x6x18 ga	6'-0"	5'-0"	4'-0"		15x6x18 ga	6'-0"	5'-0"	4'-0"
	15x6x16 ga	6'-0"	5'-0"	4'-0"		15x6x16 ga	6'-0"	5'-0"	4'-0"
	15x6x14 ga	5'-0"	4'-0"	3'-0"		15x6x14 ga	5'-0"	4'-0"	3'-0"
	15x6x12 ga	5'-0"	4'-0"	3'-0"		15x6x12 ga	5'-0"	4'-0"	3'-0"
4'-0" (R300) 40 pcf	15x6x20 ga	6'-0"	5'-0"	4'-0"	4'-0" (R300) 40 pcf	15x6x20 ga	6'-0"	5'-0"	4'-0"
	15x6x18 ga	6'-0"	5'-0"	4'-0"		15x6x18 ga	6'-0"	5'-0"	4'-0"
	15x6x16 ga	5'-0"	4'-0"	3'-0"		15x6x16 ga	5'-0"	4'-0"	3'-0"
	15x6x14 ga	5'-0"	4'-0"	3'-0"		15x6x14 ga	5'-0"	4'-0"	3'-0"
	15x6x12 ga	4'-0"	3'-0"	2'-0"		15x6x12 ga	4'-0"	3'-0"	2'-0"
4'-0" (R350) 40 pcf	15x6x20 ga	6'-0"	5'-0"	4'-0"	4'-0" (R350) 40 pcf	15x6x20 ga	6'-0"	5'-0"	4'-0"
	15x6x18 ga	6'-0"	5'-0"	4'-0"		15x6x18 ga	6'-0"	5'-0"	4'-0"
	15x6x16 ga	5'-0"	4'-0"	3'-0"		15x6x16 ga	5'-0"	4'-0"	3'-0"
	15x6x14 ga	5'-0"	4'-0"	3'-0"		15x6x14 ga	5'-0"	4'-0"	3'-0"
	15x6x12 ga	4'-0"	3'-0"	2'-0"		15x6x12 ga	4'-0"	3'-0"	2'-0"
4'-0" (R400) 40 pcf	15x6x20 ga	6'-0"	5'-0"	4'-0"	4'-0" (R400) 40 pcf	15x6x20 ga	6'-0"	5'-0"	4'-0"
	15x6x18 ga	6'-0"	5'-0"	4'-0"		15x6x18 ga	6'-0"	5'-0"	4'-0"
	15x6x16 ga	5'-0"	4'-0"	3'-0"		15x6x16 ga	5'-0"	4'-0"	3'-0"
	15x6x14 ga	5'-0"	4'-0"	3'-0"		15x6x14 ga	5'-0"	4'-0"	3'-0"
	15x6x12 ga	4'-0"	3'-0"	2'-0"		15x6x12 ga	4'-0"	3'-0"	2'-0"
4'-0" (R450) 40 pcf	15x6x20 ga	6'-0"	5'-0"	4'-0"	4'-0" (R450) 40 pcf	15x6x20 ga	6'-0"	5'-0"	4'-0"
	15x6x18 ga	6'-0"	5'-0"	4'-0"		15x6x18 ga	6'-0"	5'-0"	4'-0"
	15x6x16 ga	5'-0"	4'-0"	3'-0"		15x6x16 ga	5'-0"	4'-0"	3'-0"
	15x6x14 ga	5'-0"	4'-0"	3'-0"		15x6x14 ga	5'-0"	4'-0"	3'-0"
	15x6x12 ga	4'-0"	3'-0"	2'-0"		15x6x12 ga	4'-0"	3'-0"	2'-0"
4'-0" (R500) 40 pcf	15x6x20 ga	6'-0"	5'-0"	4'-0"	4'-0" (R500) 40 pcf	15x6x20 ga	6'-0"	5'-0"	4'-0"
	15x6x18 ga	6'-0"	5'-0"	4'-0"		15x6x18 ga	6'-0"	5'-0"	4'-0"
	15x6x16 ga	5'-0"	4'-0"	3'-0"		15x6x16 ga	5'-0"	4'-0"	3'-0"
	15x6x14 ga	5'-0"	4'-0"	3'-0"		15x6x14 ga	5'-0"	4'-0"	3'-0"
	15x6x12 ga	4'-0"	3'-0"	2'-0"		15x6x12 ga	4'-0"	3'-0"	2'-0"